R. A. PRAVEEN MALIKA WIJERATHNA

B.Sc. (Hons) in Mechanical & Process Engineering | University of Jaffna

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LinkedIn   Portfolio

Professional Summary

Graduate Mechanical Engineer with hands on industrial experience in machine design, manufacturing, process improvement, and engineering project execution within a high volume FMCG environment. Experienced in developing industrial equipment, supervising fabrication, coordinating equipment installation and commissioning, and implementing practical engineering solutions. Proficient in SolidWorks, PLC/HMI integration, engineering documentation, and cross functional collaboration, with a strong commitment to continuous improvement and operational excellence.

Education

B.Sc. (Hons) in Mechanical and Process Engineering *Apr 2022 – Jul 2026*

University of Jaffna, Sri Lanka

Professional Experience

Automation & Machine Design Engineering Intern *Nov 2025 – May 2026*

Hemas Consumer Brands – Dankotuwa, Sri Lanka

- Executed machine design, industrial automation, and manufacturing improvement projects.
- Engineered mechanical components, developed 3D models, detailed engineering drawings, and technical documentation.
- Integrated Siemens PLC and HMI control systems with mechanical equipment.
- Supervised fabrication activities carried out by outsourced contractors.
- Executed equipment installation, commissioning and commercial production trials.
- Coordinated with Production, Quality Assurance, Research & Development, Safety, and external contractors throughout machine development and implementation.
- Applied engineering methodologies including 5S, Kaizen, FMEA, vibration analysis, and engineering documentation to support continuous improvement initiatives.

Engineering Projects

Semi-Automatic Shampoo Filling Machine Development | Link

Industrial Project | Hemas Consumer Brands

- Designed and developed a semi-automatic shampoo filling machine using SolidWorks to improve production flexibility and commercial shampoo filling operations.
- Designed and assembled the electrical control panel and configured Siemens PLC/HMI systems.
- Applied Clean-in-Place (CIP) methodology within the machine design.
- Coordinated with contractors on fabrication activities and prepared job documentation.
- Supervised commercial trials with production, QA, safety, and R&D teams and implemented design modifications based on operational feedback.
- Authored change control documentation and monitored machine performance throughout commercial production.

Fully Automatic Shampoo Filling Machine Design | Link

Industrial Project | Hemas Consumer Brands

- Designed a fully automatic shampoo filling machine using SolidWorks, engineering the machine framework, bottle positioning mechanism, and filling mechanism.
- Coordinated with an overseas equipment supplier to specify and select a variable pitch system.

Cap Feeder and Elevator Machine Implementation | Link

Industrial Project | Hemas Consumer Brands

- Executed the installation and commissioning of an imported cap feeder & elevator machine.
- Designed and 3D printed interchangeable cap feeder mechanisms for different bottle types, ensuring accurate and reliable cap placement and conducted production trials.

Master Bag Packing Machine Improvement | Link

Industrial Project | Hemas Consumer Brands

- Optimized machine control logic to automatically rotate sanitary napkin packets prior to master bag packing.
- Implemented an automated packet rotation mechanism into the production transfer process.

Modelling, Development, and Performance Assessment of a Hybrid-Powered Thermoelectric Refrigerator | Link *Dec 2024 – Nov 2025*

Undergraduate Research Project

- Developed a component level simulation model using MATLAB Simulink/Simscape to predict the thermal and electrical performance of a hybrid thermoelectric refrigeration system.
- Validated model accuracy against prototype test data using statistical analysis, establishing a reliable virtual testbed for design evaluation.
- Reduced reliance on repeated physical prototyping by leveraging the simulation environment to evaluate and optimize design alternatives.
- Directed design outcomes toward a low cost, portable, energy efficient cooling solution suited for off-grid and rural applications.

Technical Skills

- **CAD & Engineering Software:** SolidWorks, EPLAN, MATLAB Simulink & Simscape, Microsoft Project
- **Industrial Automation:** Siemens PLC, Haiwell PLC, HMI Programming, Servo Drives, VFD Configuration, Pneumatic Systems
- **Mechanical Engineering:** Machine Design, Vibration Analysis, Root Cause Analysis, CIP System Design
- **Engineering Practices:** Engineering Documentation, 5S, Kaizen, FMEA

Leadership & Extracurricular Activities

- **Acting President** – Engineering Students' Union, University of Jaffna *Feb 2025 – Nov 2025*
- **Vice President** – Engineering Students' Union, University of Jaffna *Oct 2024 – Feb 2025*
- **Vice President** – Mechanical Engineering Society, University of Jaffna *2023 – 2024*
- **President** – Divisional Federation of Youth Clubs, Madulla Division, Sri Lanka Youth *2021 – 2022*

Awards & Achievements

- **Bronze Award – IESL Techno Exhibition (2023 & 2024):** Recognized by the Institution of Engineers Sri Lanka (IESL) for outstanding engineering projects; represented the University of Jaffna as Team Leader (2024) and Demonstrator (2023).

References

Eng. Disnaka Imal
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